1) $\mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} \left(2e^{-5t}u(t-5) + 3e^{-8|t|}\right) e^{-j\omega t}\, dt$

$= \int_{5}^{\infty} 2e^{-5t} e^{-j\omega t}\, dt + \int_{-\infty}^{0} 3e^{8t} e^{-j\omega t}\, dt + \int_{0}^{\infty} 3e^{-8t} e^{-j\omega t}\, dt$

$= 2\left[\frac{e^{-(5+j\omega)t}}{-(5+j\omega)}\Big|_{5}^{\infty}\right] + 3\left[\frac{e^{(8-j\omega)t}}{8-j\omega}\Big|_{-\infty}^{0}\right] + 3\left[\frac{e^{-(8+j\omega)t}}{-(8+j\omega)}\Big|_{0}^{\infty}\right]$

$= 2\,\frac{e^{-25} \cdot e^{-5j\omega}}{5+j\omega} + 3\left(\frac{1}{8-j\omega} + \frac{1}{8+j\omega}\right)$

$= 2e^{-25}\left(\frac{e^{-5j\omega}}{5+j\omega}\right) + \frac{48}{64+\omega^2}$

2) a. $\cos(\omega_0 n) = \frac{1}{2}\left(e^{j\omega_0 n} + e^{-j\omega_0 n}\right)$

$x[n] = \sum_{k=\langle N\rangle} a_k e^{jk\omega_0 n}$

$k=1$ için $a_1 = \frac{1}{2} = a_{1+N}$

$a_0 = 0$ (ortalama 0) cos için

$k=-1$ için $a_{-1} = \frac{1}{2} = a_{-1+N}$

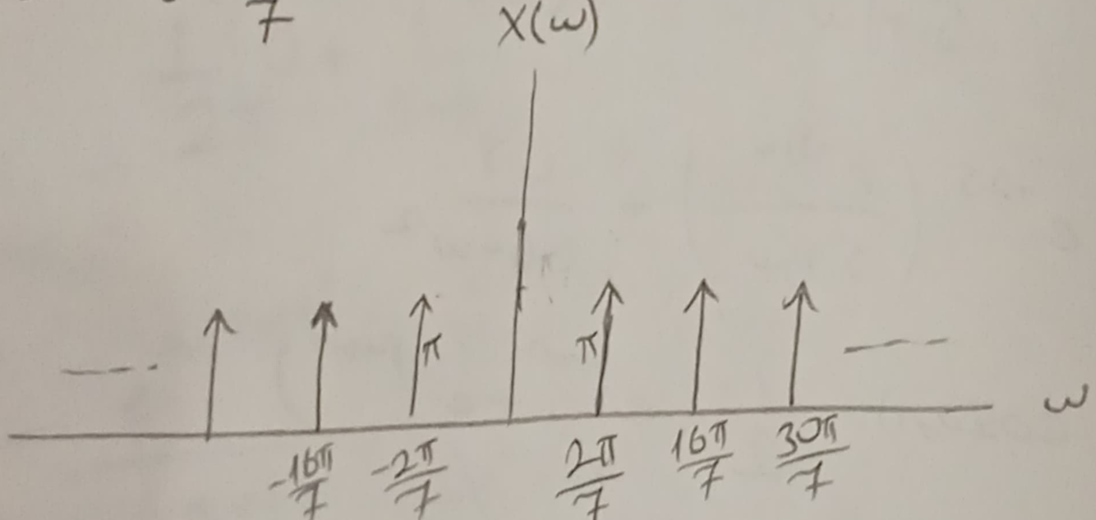
b. $\cos(\omega_0 n) \Rightarrow x[n] = \frac{1}{2}\left(e^{j\omega_0 n} + e^{-j\omega_0 n}\right)$

$$X(e^{j\omega}) = \frac{1}{2}\sum_{n=-\infty}^{\infty} e^{j\omega_0 n} \cdot e^{-j\omega n} + \frac{1}{2}\sum_{-\infty}^{\infty} e^{-j\omega_0 n} e^{-j\omega n}$$

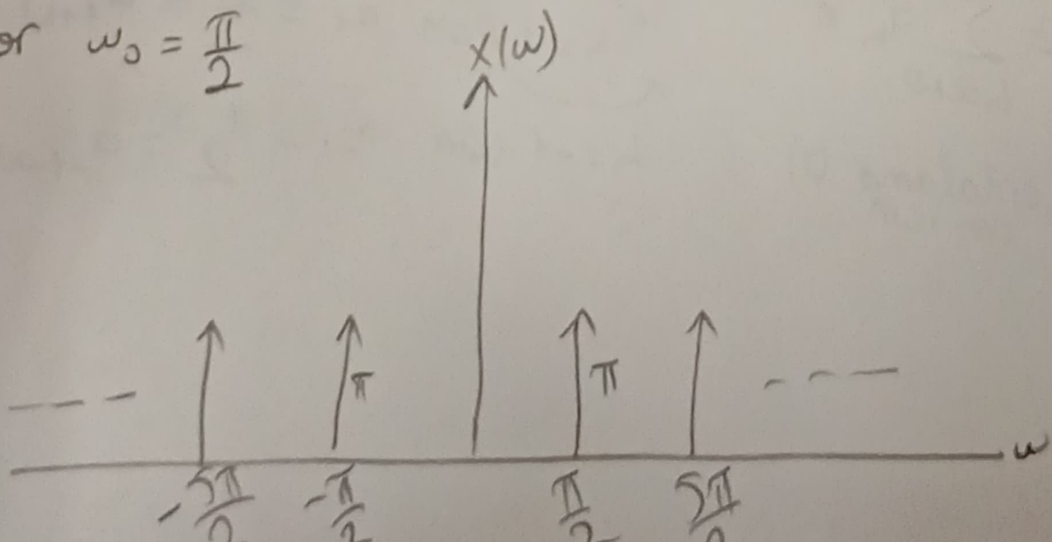
$$= \frac{1}{2}\underbrace{\sum_{n=-\infty}^{\infty} e^{-j(\omega-\omega_0)n}}_{2\pi \sum \delta(\omega-\omega_0+2\pi k)} + \frac{1}{2}\underbrace{\sum_{n=-\infty}^{\infty} e^{-j(\omega+\omega_0)n}}_{2\pi \sum \delta(\omega+\omega_0+2\pi k)}$$

$$= \pi \sum_{k=-\infty}^{\infty} \delta(\omega-\omega_0+2\pi k) + \pi \sum_{k=-\infty}^{\infty} \delta(\omega+\omega_0+2\pi k)$$

for $\omega_0 = \frac{2\pi}{7}$



for $\omega_0 = \frac{\pi}{2}$

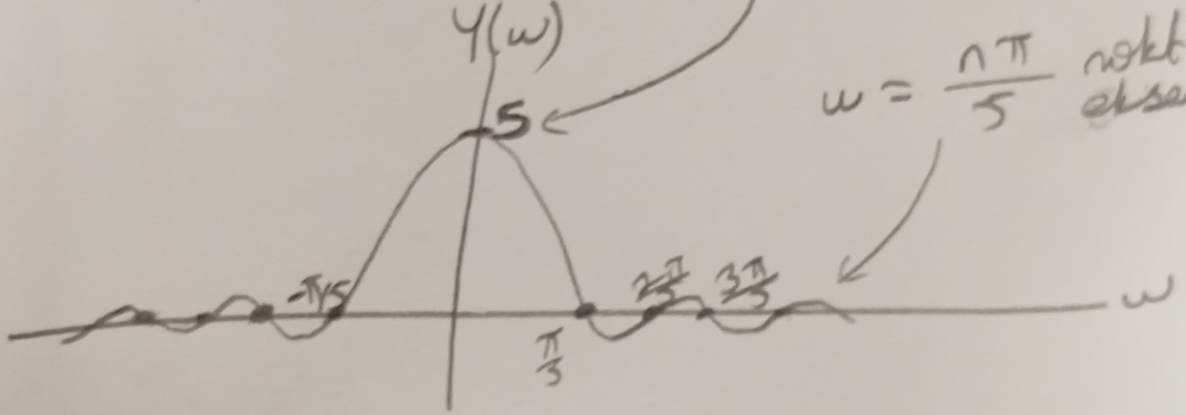


3) $$Y(\omega) = \int_{-\infty}^{\infty} y(t)\, e^{-j\omega t}\, dt = \int_{-5}^{5} \frac{1}{2}\, e^{-j\omega t}\, dt$$

$$= \frac{1}{2} \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{-5}^{5} = \frac{e^{j5\omega} - e^{-j5\omega}}{2j\omega}$$

$$Y(\omega) = \frac{\sin(5\omega)}{\omega} \cdot \frac{5}{5} = \mathrm{sinc}(5\omega) \cdot 5$$

$$\boxed{\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}}$$



$\omega = \frac{n\pi}{5}$ noktalarında ekseni keser

4) a) $$x(t)=\frac{1}{2\pi}\int_{0}^{\infty} e^{-5w}\, e^{jwt}\, dw$$

$$=\frac{1}{2\pi}\,\frac{e^{-5w+jwt}}{-5+jt}\Bigg|_{0}^{\infty} = \underline{\frac{1}{2\pi}\left(\frac{1}{5-jt}\right)}$$

b) $$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-5|w|}\, e^{jwt}\, dw \Rightarrow \underbrace{\int_{-\infty}^{0} e^{w(5+jt)}\, dw}_{\frac{1}{5+jt}} + \underbrace{\int_{0}^{\infty} e^{w(-5+jt)}\, dw}_{\frac{1}{5-jt}}$$

$$= \underline{\frac{1}{2\pi}\left(\frac{10}{25+t^2}\right)}$$